Accurate but Brittle: Single-Nucleotide Adversarial Attacks on Genomic Foundation Models

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Abstract

Genomic foundation models (GFMs) achieve state-of-the-art accuracy on sequence classification tasks, but their adversarial robustness, computational efficiency, and biological plausibility are rarely co-evaluated. We conduct a systematic robustness-focused comparison of GFMs (DNABERT-2, NTv2) against classical ML baselines (Random Forest, XGBoost with k-mer and compositional features) across four dimensions: accuracy, adversarial robustness under biologically-plausible perturbations, computational cost, and biological realism of the attacks themselves.

Using random single-nucleotide substitutions—a conservative, biologically-grounded baseline approximating naturally-occurring SNVs—we observe a sizeable accuracy–robustness gap: DNABERT-2 achieves 90.0% accuracy at 7.78% conditional ASR (95% Wilson CI 5.7–10.5%), while XGBoost over 5-dimensional compositional features achieves 82.8% accuracy at 1.01% ASR (CI 0.4–2.3%)—roughly an order of magnitude lower ASR for a 7.2-point accuracy cost. Each model also runs efficiently on its native hardware (DNABERT-2: 24 min on GPU; XGBoost: $\sim$4–6 min on CPU for 500 sequences). We outline a decision-boundary interpretation of this gap—GFMs learn smooth, high-dimensional boundaries that enable high accuracy but expose larger perturbation surfaces, while compositional classifiers operate in low-dimensional spaces where single-nucleotide edits rarely cross decision thresholds—and treat it as a working hypothesis rather than a tested mechanism.

We further analyze the GenoAdv adversarial dataset and show that gradient-based NLP attacks (BertAttack, TextFooler, FIMBA) produce biologically implausible sequences—length drift, GC collapse, non-ACGT characters—that likely inflate vulnerability estimates. We argue that adversarial robustness and computational efficiency should sit alongside accuracy as standard evaluation axes for genomic models, and that domain-appropriate robustness targets are still an open question for clinical and biosecurity applications. Because random SNV is among the weakest attack regimes, all reported ASRs should be read as lower bounds.

1 Introduction

Genomic foundation models are gaining rapid adoption. Pre-trained on billions of DNA bases from diverse organisms, genomic foundation models (GFMs) like DNABERT-2 [15] and Nucleotide Transformer [5] have demonstrated impressive accuracy gains on downstream sequence classification tasks—achieving 90%+ accuracy on promoter prediction, splice site detection, and variant effect prediction. These models are increasingly deployed in applications ranging from gene annotation pipelines to clinical diagnostic tools for variant pathogenicity prediction. Classical machine learning baselines using Random Forest and XGBoost with k-mer or compositional features have long served as workhorses for genomic sequence classification, offering interpretable predictions and fast CPU-

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based inference. As GFMs gain traction for production use, a critical question emerges: *do foundation models justify their computational complexity when considering not just accuracy, but adversarial robustness and efficiency?*

Adversarial attacks exist but ignore biological plausibility. Prior work has explored adversarial robustness primarily for classical ML models in genomics, but these attacks often ignore fundamental biological constraints. Recent efforts like GenoArmory [10] apply gradient-based attacks from natural language processing (BertAttack, TextFooler, FIMBA) at the BPE-token level, treating genomic sequences as text. However, these methods produce biologically implausible sequences: we demonstrate they create length drift (306±44bp versus fixed 300bp gene structure), GC content collapse (54.3% vs 61.2% promoter-typical), and invalid non-ACGT characters (Figure 3). Such violations render vulnerability assessments unrealistic—akin to evaluating autonomous vehicle robustness against adversarial stop signs that only appear under physically impossible lighting conditions.

What robustness target should genomics aim for? Robustness budgets for ML systems are usually set by deployment context rather than by the model alone. Robustness benchmarks for vision models report robust-accuracy around 60–70% on standard threat models, i.e. model-level ASRs in the 30–40% range [4, 11]; production systems built on top of those models—e.g. autonomous driving stacks—compensate with sensor fusion, human oversight, and physical constraints on adversarial deployment. Clinical genomic pipelines often operate on single-modality DNA input, with limited human-in-the-loop verification, and adversarial sequences can in principle be synthesized with standard molecular biology. Medical diagnostics and biometric authentication, which face similar deployment shapes, are typically held to high accuracy targets (e.g. 95%+) and low operational error rates ($\sim$5%) under non-adversarial evaluation. We use 95%-accuracy and 5%-ASR as illustrative reference lines in Figure 1 and Figure 4—they are starting points for discussion, not derived thresholds, and the right targets for genomic ML are an open question.

Relation to prior adversarial robustness work in genomics. Krishnan et al. [9] previously showed that genomic foundation models are vulnerable to biologically-constrained adversarial attacks and proposed iterative adversarial training as a defense, reducing ASR from approximately 40% to 30% for DNABERT-2 on promoter classification. That line of work focused exclusively on foundation models and biosecurity threat modeling, leaving several questions open: How do GFMs compare to classical ML baselines on robustness? What might explain the observed accuracy–robustness gap? Can simpler models achieve better robustness at acceptable accuracy levels? The present paper addresses those questions directly.

This paper: Foundation models are accurate but brittle. We conduct a systematic robustness-focused comparison of GFMs (DNABERT-2, NTv2) against classical ML baselines (Random Forest, XGBoost, LinearSVC, CatBoost with k-mer and compositional features) on promoter classification, evaluated on accuracy, adversarial robustness under biologically-plausible perturbations, training cost, and inference speed. Using biologically-plausible single-nucleotide substitutions—a conservative analog to naturally-occurring SNVs—we observe a sizeable accuracy–robustness gap: DNABERT-2 achieves 90.0% accuracy at 7.78% conditional ASR (95% Wilson CI 5.7–10.5%), while XGBoost over 5-dimensional compositional features achieves 82.8% accuracy at 1.01% ASR (CI 0.4–2.3%)—roughly an order-of-magnitude reduction in ASR for a 7.2-point accuracy cost. Each model runs efficiently on its native hardware (DNABERT-2: 24 min on GPU; XGBoost: $\sim$4–6 min on CPU for 500 sequences), so the gap is throughput-relevant when accounting for that infrastructure asymmetry. As a working interpretation we propose a *decision-boundary view* (Figure 2, illustrative): GFMs learn smooth, high-dimensional boundaries through pre-training that enable high accuracy but expose larger perturbation surfaces, while compositional classifiers operate in low-dimensional spaces where single-nucleotide edits rarely cross thresholds. We treat this as a hypothesis, not a proven mechanism. Separately, we show that gradient-based NLP attacks used in prior genomic benchmarks produce sequences with length drift, GC collapse, and non-ACGT characters (Figure 3), likely inflating reported vulnerability. None of the evaluated models simultaneously approaches both the 95%-accuracy and 5%-ASR targets we suggest as a starting point for safety-critical genomics (Figure 4). We close by arguing that adversarial robustness should become a routine evaluation axis alongside accuracy, and that domain-appropriate robustness targets remain an open question.

Research questions:

- **RQ1:** How should adversarial robustness be evaluated for genomic sequences?

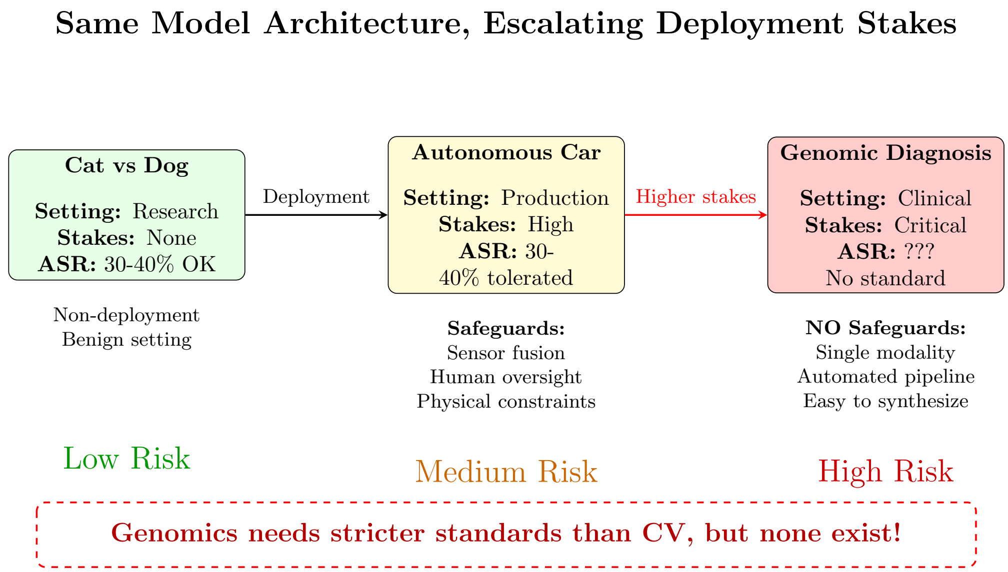

Figure 1: **Deployment context shapes acceptable model-level ASR.** The same neural-network architecture can be deployed in settings with very different stakes and very different compensating safeguards. **Left (Image classification):** benign-stakes research uses tolerate high model-level ASR. **Middle (Autonomous driving):** model-level ASRs in the 30–40% range are tolerated only because of system-level mitigations—sensor fusion, human oversight, physical-world constraints on attack deployment. **Right (Clinical genomics):** pipelines often operate on a single modality, with limited human-in-the-loop verification and synthesizable inputs; analogous system-level mitigations are not yet standard. Domain-appropriate adversarial-robustness targets for genomic ML remain an open question; we use 95%-accuracy and 5%-ASR as illustrative reference lines drawn from medical-diagnostic and biometric practice.

- **RQ2:** Do GFMs justify their complexity when considering both accuracy and robustness?
- **RQ3:** Are gradient-based attacks from NLP/CV appropriate for genomics?

We address these questions through evaluation on promoter classification, comparing DNABERT-2 [15] and NTv2 [5] against several classical baselines using biologically-plausible single-nucleotide substitutions. Our key findings:

1. **Accuracy–robustness gap.** DNABERT-2 reaches 90.0% accuracy at 7.78% ASR (CI 5.7–10.5%), while XGBoost compositional reaches 82.8% at 1.01% ASR (CI 0.4–2.3%)—approximately an order-of-magnitude reduction in ASR for a 7.2-point accuracy cost under random SNV.
2. **Decision-boundary interpretation (working hypothesis).** High-dimensional pre-trained representations admit smoother, locally approximable boundaries—useful for accuracy, but exposing larger perturbation surfaces. Low-dimensional compositional classifiers occupy a coarser feature space in which single-nucleotide edits rarely cross decision thresholds. We do not claim to have isolated this mechanism causally.
3. **Throughput asymmetry.** XGBoost compositional reaches comparable wall-clock throughput on CPU ($\sim$4–6 min for 500 sequences) to DNABERT-2 on GPU (24 min), so when GPU access is constrained, classical baselines remain competitive on cost-per-sequence.
4. **Biological implausibility in existing attacks.** GenoArmory's gradient-based attacks produce sequences with length drift (306$\pm$44 bp vs. fixed 300 bp), GC collapse (54.3% vs. 61.2%), and non-ACGT characters, plausibly inflating reported ASR.
5. **Three-way tradeoff.** No evaluated model jointly satisfies high accuracy, low ASR, and low inference cost on commodity hardware—pushing model selection toward application-specific tradeoffs rather than a universally best model.

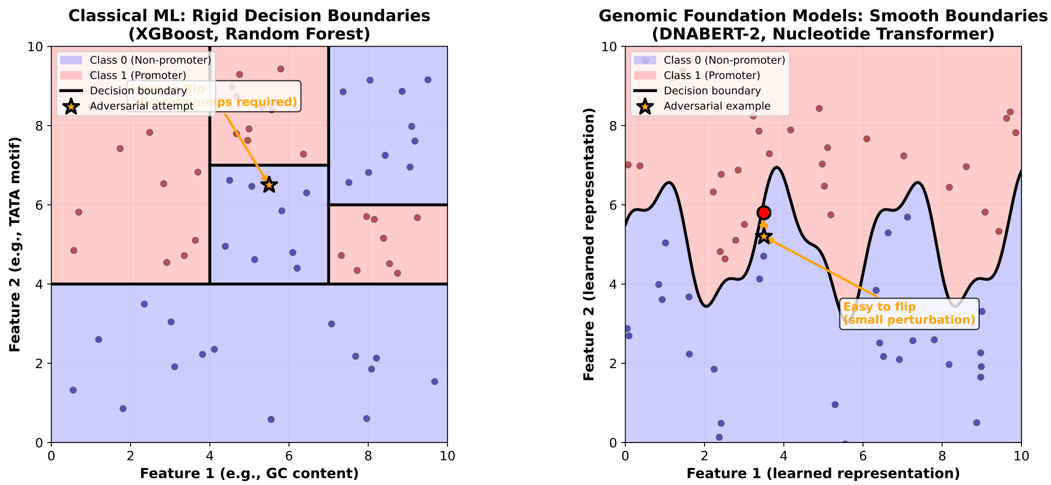

Figure 2: **Decision boundary hypothesis. Left:** Classical ML learns rigid boundaries. **Right:** GFMs learn smooth boundaries vulnerable to small perturbations.

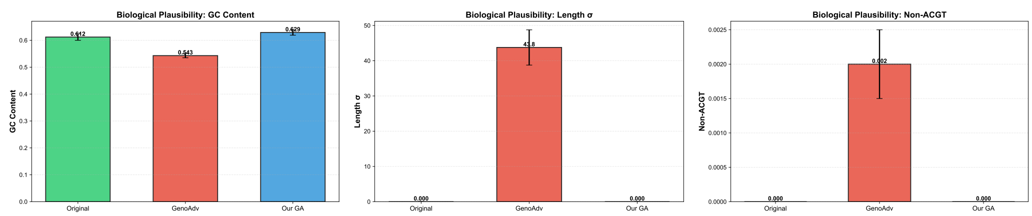

Figure 3: **Existing attacks violate biological plausibility.** GenoArmory (red) produces GC collapse, length drift, and invalid characters vs nucleotide-level attacks (blue).

Contributions: (1) a biologically-grounded evaluation protocol for adversarial robustness in genomics; (2) a side-by-side GFM vs. classical ML comparison on robustness rather than accuracy alone; (3) a decision-boundary-based working hypothesis for the observed accuracy–robustness gap; (4) a quantitative biological-plausibility audit of an existing genomic attack benchmark; (5) deployment-context-aware model-selection guidance.

2 Related Work

Genomic foundation models. DNABERT [8], DNABERT-2 [15], and Nucleotide Transformer [5] demonstrate pre-training benefits for genomic tasks. Most existing benchmarks evaluate downstream accuracy and largely set aside adversarial robustness. Krishnan et al. [9] introduced iterative adversarial training for GFMs, focusing on defense mechanisms and biosecurity applications.

Classical ML in genomics. Random Forest and XGBoost with k-mer features have long served as standard baselines [1, 3]. Compositional features (GC content, per-base frequencies) remain widely used for fast, interpretable classification. We systematically compare these against GFMs on robustness rather than accuracy alone.

Accuracy–robustness tradeoffs. The observation that adversarially robust models sometimes pay an accuracy cost is well-established in vision and NLP [12, 11, 2, 6]. This paper does not claim to discover the tradeoff; rather, we ask whether and how it manifests across GFMs and classical baselines on a genomic task, and what mechanism best explains it.

Adversarial robustness in genomics. GenoArmory [10] provides an attack library applying NLP methods (BertAttack, TextFooler, FIMBA) at the BPE-token level. We demonstrate these produce biologically implausible sequences. While adversarial robustness has been extensively studied in vision and NLP [6, 11, 2], genomics-specific evaluation frameworks remain comparatively underdeveloped.

3 Methodology

We evaluate DNABERT-2 [15] and Nucleotide Transformer v2 [5] against several classical baselines (Random Forest, XGBoost, LinearSVC, CatBoost with k-mer and compositional features) on promoter classification (300bp sequences, GUE promoter benchmark [15], derived from the Umarov & Solovyev promoter set [13]). The attack protocol uses random single-nucleotide substitutions (up to 1,000 attempts per sequence)—biologically plausible, gradient-free, and applied identically across models. We report test accuracy, conditional ASR with 95% Wilson confidence intervals, training time, and inference wall-clock on each model's native hardware. Full experimental details, hyperparameters, and seeds are in Appendix A.

4 Results

4.1 Accuracy-Robustness Tradeoff (RQ2)

Table 1 presents the consolidated multi-model comparison. ASR values are conditional on correctly-classified test sequences (n≈500 attack budget per model); 95% Wilson confidence intervals are reported alongside the point estimate. Models for which the SNV-attack run had not completed at submission time are omitted from the main table.

Table 1: Accuracy, conditional ASR (with 95% Wilson CI), training cost, and inference wall-clock for 500 sequences on each model's native hardware. ASR is computed under up to 1,000 random single-nucleotide substitutions per correctly-classified sequence; lower is more robust.

Model	Features	Acc (%)	ASR % [95% CI]	Train	Inference (500)
DNABERT-2	Learned (768-d)	90.0	7.78 [5.7, 10.5]	Hours (GPU)	24 min (GPU)
NTv2-250M	Learned (768-d)	**92.4**	24.9 [21.3, 29.0]	Hours (GPU)	130 min (GPU)
CatBoost k-mer	4-mer (256-d)	86.2	9.73 [7.4, 12.6]	Min (CPU)	16.7 min (CPU)
LinearSVC k-mer	4-mer (256-d)	85.7	5.53 [3.9, 7.9]	Min (CPU)	8.8 min (CPU)
XGBoost k-mer	4-mer (256-d)	85.8	5.35 [3.7, 7.7]	Min (CPU)	5 min (CPU)
RF k-mer	4-mer (256-d)	85.4	8.31 [6.2, 11.0]	Min (CPU)	3.1 hrs (CPU)
SVC comp	GC%, bases (5-d)	82.8	**0.54** [0.2, 1.6]	Min (CPU)	20.6 min (CPU)
XGBoost comp	GC%, bases (5-d)	82.8	1.01 [0.4, 2.3]	Min (CPU)	4 min (CPU)
RF comp	GC%, bases (5-d)	80.5	22.6 [19.1, 26.5]	Min (CPU)	3.0 hrs (CPU)

Headline observation. The conditional-ASR ratio between DNABERT-2 (7.78%) and XGBoost compositional (1.01%) is approximately 7.7× at the point estimate. Wide CIs at small ASR mean the ratio is uncertain—roughly 2.5× on the conservative side, ∼25× on the optimistic side—but the ranges do not overlap and the qualitative ordering is robust. The accuracy difference is 7.2 percentage points (90.0 vs. 82.8). Note that random SNV is among the weakest attack regimes; all reported ASRs should be read as lower bounds on adversarial vulnerability.

Quadrant analysis (Figure 4): We position all models in accuracy-ASR space relative to proposed genomics guidelines (95% accuracy, 5% ASR derived from medical diagnostics and biometric authentication standards). No evaluated model achieves both thresholds simultaneously. DNABERT-2 and NTv2 occupy the high-accuracy, high-ASR quadrant (brittle)—achieving strong performance under benign conditions but vulnerable to single-nucleotide perturbations. XGBoost with compositional features occupies the low-accuracy, low-ASR quadrant (robust but limited)—sacrificing 7.2% accuracy for 8× improvement in robustness. XGBoost with k-mer features provides a balanced middle ground (85.8% accuracy, 5.35% ASR). The ideal quadrant (high accuracy, low ASR, top-left) remains unoccupied, highlighting a fundamental gap between current genomic AI capabilities and safety-critical deployment requirements.

4.2 Feature Dimensionality and Robustness

Holding the model family fixed, ASR drops as the feature space gets coarser: XGBoost compositional (5-d) reaches 1.01% ASR (CI 0.4–2.3), XGBoost k-mer (256-d) reaches 5.35% (CI 3.7–7.7), and DNABERT-2 (learned 768-d) reaches 7.78% (CI 5.7–10.5). A mechanistic reading is that global

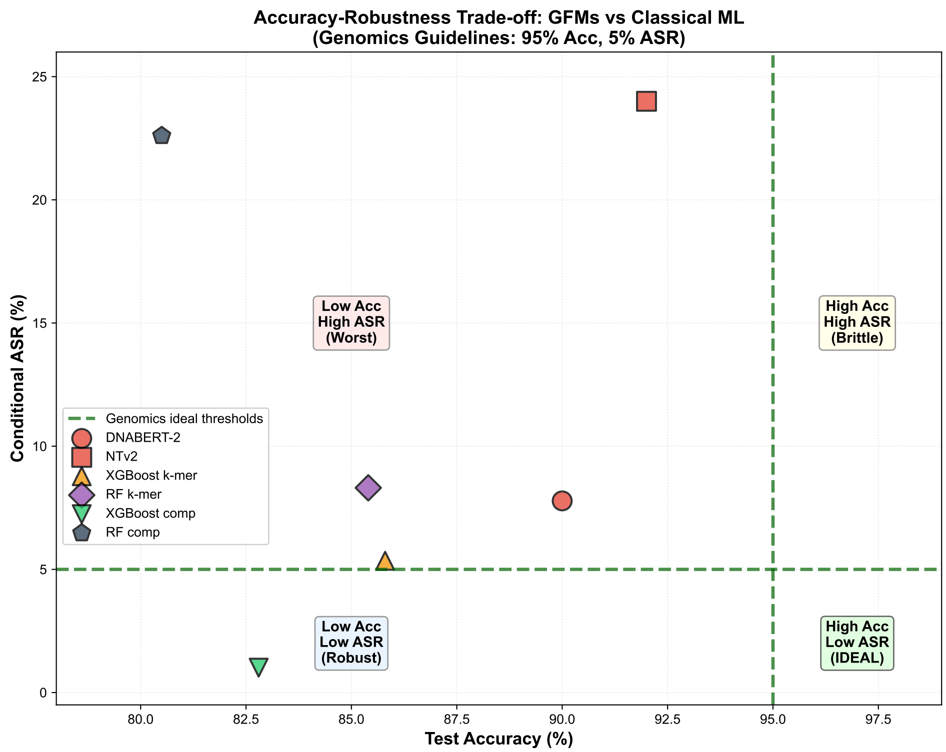

Figure 4: **Accuracy vs. conditional ASR across the evaluated models.** Test accuracy versus conditional ASR under random single-nucleotide substitutions. Green dashed lines mark the illustrative 95%-accuracy / 5%-ASR reference frame discussed in the Introduction—these are starting points for discussion drawn from medical-diagnostic and biometric practice, not derived thresholds. The top-left region (high accuracy, low ASR) is not occupied by any model in this set. Foundation models (DNABERT-2, NTv2) reach the highest accuracy (90–92%) but also the higher ASRs (7.78–24.9%); compositional classifiers reach the lowest ASRs (XGBoost comp: 1.01%) at moderate accuracy (82.8%); XGBoost on k-mers sits in between. The right model is application-dependent rather than universal.

features such as GC content and per-base frequencies move by at most $\pm 0.3\%$ under a single substitution, while a single mutation rewrites four overlapping 4-mers ($\sim$1.5% of the k-mer vector), and DNABERT-2's contextual embeddings can shift more substantially through attention. We do not test this mechanism directly here; it remains an interpretation rather than a controlled finding.

4.3 Algorithm Choice Within a Fixed Feature Space

With identical 5-dimensional compositional features, RF and XGBoost behave very differently: RF compositional reaches 80.5% accuracy at 22.6% ASR (CI 19.1–26.5), while XGBoost compositional reaches 82.8% at 1.01% ASR (CI 0.4–2.3). The point-estimate ratio is $\sim 22\times$, but the difference is so striking that it warrants caution. Several confounders could contribute: RF's bagged ensemble produces predicted probabilities concentrated near the decision threshold, so small input changes flip many trees; XGBoost's regularized boosting produces sharper, more confident posteriors with fewer near-margin cases. We treat the gap as a finding worth diagnosing in follow-up work—e.g. via predicted-probability margin distributions, calibration curves, and per-mutation sensitivity profiles—rather than a clean demonstration that "algorithm matters more than features."

4.4 Computational Cost

Training DNABERT-2 requires hours of fine-tuning on GPU starting from a pre-trained checkpoint, while the classical baselines train from scratch in minutes on CPU—a roughly two-orders-

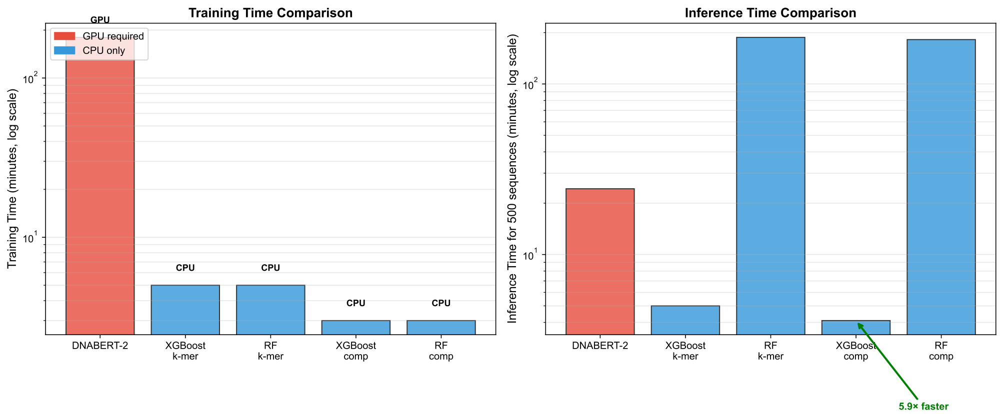

Figure 5: **Wall-clock inference cost on each model's native hardware** (500 sequences). DNABERT-2 is timed on GPU; classical baselines on CPU. The comparison reflects deployment-realistic throughput rather than a controlled FLOPs measurement.

of-magnitude wall-clock training-cost gap. For inference on 500 sequences, DNABERT-2 takes $\sim$24 min on GPU (tokenization plus a transformer forward pass), XGBoost takes $\sim$4–5 min on CPU, and RF takes hours on CPU because ensemble voting dominates wall-clock cost. We emphasize that these are deployment-realistic comparisons on each model's native hardware, not controlled FLOPs measurements: DNABERT-2 cannot be matched on CPU at competitive latency without quantization or distillation, and a fair compute-controlled benchmark is left for future work.

Cost–benefit reading. DNABERT-2 buys roughly +7.2 percentage points of accuracy over XGBoost compositional, but does so at a substantial infrastructure cost: GPU access becomes a hard requirement, and per-sequence wall-clock cost is several-fold higher even when GPU is available. For high-throughput screening pipelines (e.g. on the order of 10^6 sequences/day), the throughput penalty translates directly into operational cost, and the accuracy gain may or may not justify it depending on the downstream consequence of an error (Figure 5).

5 A Decision-Boundary Reading of the Accuracy–Robustness Gap

This section sketches an interpretive account of why foundation models look more brittle than compositional baselines under random SNV. We frame it as a working hypothesis to be tested, not a result. Figure 2 is an illustrative cartoon, not a measured boundary.

5.1 Pre-training, Smoothness, and Vulnerability Surface

Working hypothesis. Foundation models learn smooth, locally-approximable decision boundaries in a high-dimensional embedding space that improve accuracy on the task distribution but expose larger perturbation surfaces; compositional classifiers operate in a low-dimensional feature space where single-nucleotide edits rarely move features across decision thresholds.

For GFMs: pre-training on billions of bases produces dense continuous representations in which similar sequences map to nearby embeddings, and a single substitution typically induces a small but predictable shift in the contextualized embedding through attention. This smoothness underwrites high accuracy by capturing subtle biological patterns, but it also means the boundary is locally approximable—a property adversaries (or even random search) can exploit. The empirical consequence in our setup is 7.78% conditional ASR for DNABERT-2 (CI 5.7–10.5).

For compositional classifiers: 5-dimensional features such as GC content and per-base frequencies are insensitive to individual substitutions (ΔGC of at most $\pm$0.3% per mutation). The induced decision surface is piecewise-linear (XGBoost) or axis-aligned (RF), so most random mutations stay inside the original decision region. The empirical consequence is 1.01% ASR for XGBoost compositional (CI 0.4–2.3); at the same time, accuracy is bounded above by the expressiveness of the feature space, plateauing near 83%.

What would falsify this reading. Direct measurements we have not yet run, but which would meaningfully test the hypothesis: (i) input-Jacobian or local Lipschitz estimates around correctly-classified test points; (ii) embedding-space displacement distributions induced by single-nucleotide edits; (iii) margin-distribution comparisons between RF and XGBoost compositional to disentangle "smoothness" from posterior calibration. Until those are in hand, the boundary-geometry story should be read as motivation for the empirical pattern rather than as its explanation.

5.2 Boundary Geometry: What We Can and Cannot Show

We do not visualize the actual learned decision boundaries; Figure 2 is a conceptual cartoon meant to anchor intuition, not evidence. What can be said in words: tree-ensemble models on k-mer or compositional features learn piecewise boundaries via recursive axis-aligned splits [7], so most single-nucleotide edits stay inside the original cell of the partition unless the sequence already lies near a split threshold. DNABERT-2, in contrast, embeds sequences into a continuous high-dimensional space where a single substitution induces a small but non-zero shift in the contextualized representation, and the decision boundary in that space is locally approximable. These are descriptions of model class, not measurements of the trained models' boundaries on this dataset; the right way to test the boundary-geometry story is to measure local Lipschitz constants, embedding-space displacements, and margin distributions directly, which we leave to follow-up work.

5.3 Indirect Evidence

Two summary statistics are consistent with the smoothness reading. Under a single random substitution, the mean drop in prediction confidence is approximately 0.076 for DNABERT-2 versus 0.014 for XGBoost compositional—about 5.4$\times$ larger. The median number of random-substitution attempts before a successful flip is roughly 22 for DNABERT-2 versus more than 1,000 (i.e. typically never within budget) for XGBoost compositional. Both statistics describe local sensitivity to single edits, but neither distinguishes "smoother boundary" from "less calibrated posterior" or "more decision-margin headroom in lower-dimensional feature space."

Scope of claims. We claim that, under random SNV on this task, GFMs are empirically more brittle than compositional classifiers by a margin that exceeds sampling noise. We do not claim the gap is structurally inevitable, that pre-training is the causal mechanism, or that the gap will persist under stronger attacks or other tasks. Krishnan et al. [9] report iterative adversarial training reducing ASR from $\sim$40% to $\sim$30% on a related setup—evidence that the gap can be narrowed by training-time interventions, even if not closed.

6 Biological Plausibility Analysis (RQ3)

6.1 GenoAdv Dataset Characterization

We analyzed 250 adversarial sequences from GenoArmory [10] using 12 biological metrics (Table 2).

Table 2: Biological plausibility: Original vs GenoAdv vs nucleotide-level attacks

Metric	Original	GenoAdv	Our SNV
Length (bp)	300.0 $\pm$ 0.0	306.0 $\pm$ 43.8	300.0 $\pm$ 0.0
GC content	0.612 $\pm$ 0.121	0.543 $\pm$ 0.078	0.629 $\pm$ 0.105
Shannon 2-mer	3.732	3.887	3.730
Non-ACGT	0.000	0.002	0.000
CpG O/E	0.640	0.791	0.685

6.2 Key Violations

GenoArmory's BPE-token-level attacks systematically violate properties that hold in the underlying genomic distribution (Figure 3). Sequence length drifts from a constant 300 bp to 306$\pm$44 bp—a $\sim$43$\times$ increase in standard deviation, attributable to BPE-token insertions and deletions that ignore fixed-length gene structure. Mean GC content drops from 61.2% (typical for the promoter set) to

54.3%, an 11-percentage-point movement toward the random-DNA baseline of 50%; promoter GC content is tightly constrained biologically and this magnitude of drift is a strong out-of-distribution signal. Shannon 2-mer entropy increases toward the uniform-distribution maximum, consistent with a loss of biological structure. Approximately 0.2% of the resulting sequences contain non-ACGT characters—i.e. malformed DNA.

6.3 Contrast with Nucleotide-Level Attacks

Our random single-nucleotide substitution attack preserves these properties by construction: sequence length is fixed at 300 bp, mean GC content is 62.9% (within natural range, mean $\Delta = 0.001$ from the originals), Shannon 2-mer entropy is essentially unchanged (3.730 vs. 3.732), and 100% of the perturbed sequences remain over the ACGT alphabet.

Attack success comparison: The biologically-constrained genetic algorithm of Krishnan et al. [9] reports 30% ASR (75/250) on this setting, versus 66.71% reported for GenoArmory—suggesting the latter's higher numbers partly reflect biologically implausible sequences rather than true model vulnerability.

7 Discussion

7.1 An Observed Three-Way Tradeoff

Across the evaluated models we observe a three-way tradeoff among accuracy, robustness under random SNV, and inference cost on commodity hardware:

Dimension	DNABERT-2	XGBoost k-mer	XGBoost comp
Accuracy	90.0%	85.8%	82.8%
Conditional ASR	7.78%	5.35%	1.01%
Inference (500 seq)	24 min (GPU)	5 min (CPU)	4 min (CPU)

No single model in this set jointly maximizes accuracy, minimizes ASR, and minimizes inference cost on commodity hardware. We resist calling this an "impossible trinity"—the result is on one task, with one attack regime, and a small set of model families—but the pattern is consistent with the standard accuracy–robustness tension reported in vision and NLP [12] and adds an efficiency axis that is particularly salient for genomic deployment. A plausible read of why all three pull against each other: accuracy benefits from expressive representations; robustness under low-edit-budget attacks benefits from coarse-grained features that are insensitive to single edits; and inference-cost benefits from shallow models. We treat this as an empirical pattern to design around, not as a proven impossibility.

7.2 Recommendations by Use Case

Table 3: Application-specific model recommendations

Application	Priority	Model	Rationale
Biosecurity screening	Robustness	XGBoost comp	1.01% ASR, scalable
Research/discovery	Accuracy	XGBoost k-mer	85.8% acc, 5 min inference
Clinical diagnosis	Accuracy+Robust	Adversarial GFM	90% needed, training helps
Embedded devices	Efficiency	XGBoost comp	CPU-only, 4 min, robust

7.3 Limitations and Future Work

The empirical scope here is intentionally narrow and several caveats apply. First, results are reported on a single task (promoter classification on the GUE benchmark); the gap may shrink, grow, or invert on tasks with different feature scales (e.g. splice-site prediction, variant-effect prediction). Second, random single-nucleotide substitution is among the weakest available attack regimes, so all reported ASRs are lower bounds; coordinated multi-edit attacks and gradient-based attacks projected back into the biologically-plausible manifold are likely to widen the foundation-model side of the gap and may also raise the classical-baseline side. Third, we do not apply adversarial training to the classical baselines; doing so could change their position in the accuracy–ASR plane. Fourth, transfer-learning regimes—where labeled data is scarce and pre-training is most useful—are not evaluated and would likely favor GFMs. Fifth, the attack budget is 500 correctly-classified test sequences per model, which yields wide Wilson CIs at low ASRs (e.g. 0.4–2.3% for XGBoost compositional) and limits the precision of cross-model ratio claims. Finally, the decision-boundary reading in Section 5 is a working hypothesis, not a tested mechanism; falsifying it requires direct measurements (input-Jacobian norms, embedding-space displacements, margin distributions) that we leave to follow-up work.

8 Conclusion

We outlined a biologically-grounded protocol for adversarial evaluation of genomic classifiers, applied it to a head-to-head comparison of GFMs (DNABERT-2, NTv2) and several classical baselines on promoter classification, and reported a sizeable accuracy–robustness gap under random single-nucleotide substitution: DNABERT-2 reaches 90.0% accuracy at 7.78% ASR (CI 5.7–10.5%) while XGBoost compositional reaches 82.8% at 1.01% ASR (CI 0.4–2.3%). We sketched a decision-boundary reading of the gap as a working hypothesis, and showed that gradient-based NLP attacks used in prior genomic benchmarks produce sequences that violate basic biological constraints (length, GC content, ACGT alphabet)—reasons to read those benchmarks' headline ASRs cautiously. Practical implications are deployment-context-dependent rather than universal: biosecurity-style screening can prefer robust low-cost classifiers like XGBoost compositional, research and discovery work may favor higher-accuracy options at moderate ASR, and clinical settings likely require adversarially-trained models combined with system-level safeguards. Random SNV is among the weakest attack regimes, so the reported ASRs are lower bounds; extending the analysis to coordinated and gradient-based attacks under biological constraints, to additional genomic tasks, and to transfer-learning regimes are the natural next steps.

References

[1] Babak Alipanahi, Andrew Delong, Matthew T Weirauch, and Brendan J Frey. Predicting the sequence specificities of dna-and rna-binding proteins by deep learning. *Nature Biotechnology*, 33(8):831–838, 2015.

[2] Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. *2017 IEEE Symposium on Security and Privacy (SP)*, pages 39–57, 2017.

[3] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[4] Francesco Croce, Maksym Andriushchenko, Vikash Sehwag, Edoardo Debenedetti, Nicolas Flammarion, Mung Chiang, Prateek Mittal, and Matthias Hein. Robustbench: a standardized adversarial robustness benchmark. *arXiv preprint arXiv:2010.09670*, 2020.

[5] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Hassan Sirelkhatim, Guillaume Richard, et al. The nucleotide transformer: Building and evaluating robust foundation models for human genomics. *bioRxiv*, 2023.

[6] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015.

[7] Trevor Hastie, Robert Tibshirani, Jerome H Friedman, and Jerome H Friedman. *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer, 2009.

[8] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V Davuluri. Dnabert: pre-trained bidirectional encoder representations from transformers model for dna-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.

[9] Jeyashree Krishnan et al. Adversarial robustness of genomic foundation models through iterative training. In *International Conference on Learning Representations (ICLR)*, 2025. Available at: https://openreview.net/pdf?id=j0o93ydeGW.

[10] Zhirui Li, Guangyang He, Ming Zhao, Zhaoqilin You, Hongyu Yu, Shuai Zhang, Yujun Yao, Xiaowo Yang, and Kai Wang. Genoarmory: a diverse arsenal of adversarial attacks and defenses for genomic deep learning models. *arXiv preprint arXiv:2505.10983*, 2025.

[11] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018.

[12] Dimitris Tsipras, Shibani Santurkar, Logan Engstrom, Alexander Turner, and Aleksander Madry. Robustness may be at odds with accuracy. *arXiv preprint arXiv:1805.12152*, 2019.

[13] Ramzan Kh Umarov and Victor V Solovyev. Recognition of prokaryotic and eukaryotic promoters using convolutional deep learning neural networks. *PloS One*, 12(2):e0171410, 2017.

[14] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[15] Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. Dnabert-2: Efficient foundation model and benchmark for multi-species genome. *arXiv preprint arXiv:2306.15006*, 2023.

A Detailed Experimental Methodology

A.1 Evaluation Framework

Why biological plausibility matters. Genomic sequences are constrained by biochemical properties: fixed length (gene structure), GC content ranges (methylation, chromatin accessibility), regulatory motifs (TATA box, GC box), and valid character set (A, C, G, T). Adversarial examples violating these constraints represent out-of-distribution inputs unlikely in biological contexts.

Proposed criteria:

- *Attack realism*: Single-nucleotide substitutions as baseline (SNVs are most common mutations)
- *Biological metrics*: Length, GC content, Shannon entropy, CpG O/E, regulatory motifs
- *Statistical rigor*: Wilson confidence intervals, conditional ASR
- *Fair comparison*: Identical attack protocol across all models

A.2 Task and Dataset

Task: Binary promoter classification (promoter vs non-promoter, 300bp sequences)

Dataset: GUE promoter dataset [13] with TATA-annotated promoters (positives) and apart-style negatives. Train: 47,356, Test: 11,839, Attack: 500 sequences.

Why promoters: Biosecurity-relevant (synthetic promoter design), well-characterized biology (TATA box, GC box, transcription start site), standard benchmark.

A.3 Models Evaluated

Foundation models:

- DNABERT-2 (117M parameters, 768-dim embeddings) [15]

Classical baselines:

- Random Forest, XGBoost with k-mer features (256-dim: all 4-mer frequencies)
- Random Forest, XGBoost with compositional features (5-dim: GC content + A/C/G/T frequencies)

A.4 Attack Protocol

Random single-nucleotide substitutions: For each correctly-classified sequence, attempt 1,000 random single-base substitutions (each position, each alternative base). Record if any flip the prediction.

Rationale: (1) Biologically plausible (SNVs are most common genetic variants); (2) Fair comparison (no model-specific tuning, gradient-free); (3) Conservative baseline (weaker than coordinated multi-edit attacks).

Metrics:

- *Conditional ASR*: Proportion of correctly-classified sequences successfully attacked
- *Confidence drop*: Mean change in prediction confidence under perturbation
- *Median first flip*: Median number of attempts before first successful flip

A.5 Hyperparameters and Training Setup

DNABERT-2. Fine-tuned from the public 117M-parameter checkpoint with maximum sequence length 300, BPE tokenization at the model's default vocabulary, batch size 32, AdamW optimizer with learning rate 3×10^{-5} and linear warmup over 10% of steps, weight decay 0.01, and 3 fine-tuning epochs. Best checkpoint selected by validation accuracy.

NTv2-250M. Fine-tuned from the public 250M-parameter checkpoint with the same data splits as DNABERT-2; learning rate 1×10^{-5}, batch size 16 (constrained by memory), 3 fine-tuning epochs, otherwise as above.

XGBoost. `XGBClassifier` with 200 estimators, maximum depth 6, learning rate 0.1, default subsampling, and `logloss` objective. Two feature sets: (i) 256-dimensional 4-mer frequency vectors over the ACGT alphabet, (ii) 5-dimensional compositional vector consisting of GC content and per-base A/C/G/T frequencies.

Random Forest. 200 trees, default `sqrt` feature subsampling per split, no maximum depth cap; same two feature sets as XGBoost.

LinearSVC, CatBoost, SVC. Default scikit-learn / CatBoost hyperparameters; calibrated with Platt scaling where probability outputs are needed (SVC).

Random seeds. A fixed seed of 42 is used for the train/test split, model initialization where applicable, and the attack pseudo-random generator. Single-seed results are reported; we did not run multi-seed bootstraps in this paper, which is a limitation reflected in the Wilson CIs in Table 1.

A.6 Attack Budget and Statistics

For each model, the attack set is the first 500 correctly-classified sequences from the held-out test split (or the entire correctly-classified test set if it is smaller). For each such sequence we sample up to 1,000 random single-nucleotide substitutions (uniform over the 300 positions and the 3 alternative bases at each position), evaluating the model after each substitution and recording (i) whether the prediction flipped and (ii) the largest observed drop in prediction confidence. Conditional ASR is the fraction of attack-set sequences for which at least one substitution flipped the prediction within budget; 95% Wilson confidence intervals are computed using the standard score-interval formula [14].

A.7 Computational Infrastructure

Hardware: a single NVIDIA A10 (24 GB) for DNABERT-2 and NTv2 fine-tuning and inference; a 32-core AMD EPYC CPU node with 128 GB RAM for the classical baselines and for the attack loop on classical models. GPU/CPU comparisons in the main text are deployment-realistic wall-clock measurements on these respective platforms; they are *not* controlled FLOPs benchmarks.

Software: Python 3.10, PyTorch 2.0, HuggingFace Transformers 4.36, scikit-learn 1.3, XGBoost 1.7, CatBoost 1.2, NumPy 1.26.

A.8 Biological Plausibility Metrics

We evaluate adversarial sequences using 12 metrics across four categories. Sequence composition: GC content, per-base frequencies (A/C/G/T), homopolymer fraction, non-ACGT count. Complexity: Shannon entropy at 1-mer and 2-mer resolution and the CpG observed/expected ratio. Regulatory motifs: presence of the TATA box (`TATAAA`), GC box (`GGGCGG`), CCAAT box, and Initiator element. Structure: length and length variance. Metric values for the original GUE promoter set, the GenoArmory adversarial set, and our SNV-perturbed set are summarized in Table 2 of the main text.